

Computer Fundamentals & Digital Skills Course

Introduction

This course has been organized in collaboration with the **Chahardah Masoom (A.S.) Cultural and Educational Center** and is instructed by **Amirhossein Asadpour**.

The purpose of this course is to develop students' practical computer skills and improve their digital literacy. Upon completing the course, participants will be able to use computers and digital technologies more confidently in their daily lives and perform common digital tasks more efficiently.

This course does not require any prior knowledge or experience.

The primary audience of this course consists of students between **12 and 15 years of age**. Experience has shown that many students, despite attending various computer-related courses, still lack fundamental computer skills. One possible reason is that many existing programs focus primarily on office applications—especially Microsoft Office—without establishing a solid understanding of computer fundamentals and practical digital skills.

This course has been designed to provide high-quality education at the lowest possible cost while starting from the absolute basics and helping students acquire the ability to:

- Work independently with an operating system.
 - Become familiar with common operating systems and understand their basic differences.
 - Perform basic office-related tasks and create simple presentations using common productivity software.
 - Identify the main hardware components of a computer, understand their functions, and recognize how they interact.
 - Become familiar with malware, computer viruses, Trojans, and ransomware, and learn methods for preventing them.
 - Protect personal accounts and sensitive information against cyber threats, including social engineering and other common attacks.
 - Become familiar with modern Artificial Intelligence models and tools and use them effectively, responsibly, and productively in everyday activities.
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Course Objectives

- Promote safe and responsible computer usage.
- Improve students' digital literacy.
- Build a solid understanding of fundamental Information Technology concepts.
- Introduce modern technologies such as Artificial Intelligence.
- Encourage students to begin learning programming.
- Provide accessible learning opportunities for people from all backgrounds.

Course Outline

1. Introduction

- Course introduction
- What is a computer?
- Input and output devices
- Basic hardware concepts
- Powering on/off a computer and introduction to software

2. Operating Systems

- Introduction to the Windows environment
- Files, folders, file paths, file formats, and their purposes
- File compression and archive extraction
- Installing and uninstalling software
- Productivity tips and useful keyboard shortcuts

3. Internet & Digital Security

- Search engines and effective web searching
- Identifying trustworthy websites
- Computer viruses
- Cyber threats and protection methods
- Password management and account security
- Hacking and social engineering awareness

4. Office Applications

- What is an Office suite?
- Microsoft Word
- Microsoft PowerPoint
- Microsoft Excel
- Introduction to useful productivity tools, including AI models and alternative office suites such as LibreOffice

5. Artificial Intelligence

- Why Artificial Intelligence?

- Using AI tools such as ChatGPT
- Capabilities of different AI models
- Strengths and limitations of AI systems
- Fundamentals of prompt engineering

6. Introduction to Programming (Python)

- What is programming and why learn it?
 - Types of programming languages
 - Introduction to Python
 - Installing the required tools
 - Writing the first program
 - Basic programming concepts and syntax
 - Developing several simple practical programs
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